

Retrieval-Augmented Intelligent Legal Assistant for Court Judgments: A Hybrid Retrieval and Evidence-Grounded Generation Approach

A. Sahasra, K. P. Shreya, V. Gowrie, K. Ishta

Department of Computer Science / Applied Machine Learning for Text Analytics

ABSTRACT

Court judgments are lengthy, citation-dense, and written in domain-specific legal language, which makes keyword-based search prone to missing semantically related precedents and makes unconstrained large language models (LLMs) prone to producing unsupported or hallucinated legal claims. This paper proposes a Retrieval-Augmented Generation (RAG) system for court-judgment question answering that combines lexical retrieval (BM25), dense semantic embeddings, metadata filtering, and cross-encoder reranking to build a hybrid retrieval pipeline, followed by legal-structure-aware chunking and an evidence-first generation stage in which every material claim in the generated answer is required to be traceable to a retrieved passage. A verification module performs a faithfulness check and triggers a low-confidence "insufficient evidence" response when supporting evidence is weak, rather than allowing the system to answer regardless of relevance. We describe the proposed architecture, the dataset and preprocessing strategy, the mathematical formulation of the hybrid scoring and evaluation metrics, and a five-dimension evaluation protocol spanning retrieval quality (Recall@k, Precision@k, MRR/nDCG), answer quality, faithfulness, usability (latency, citation coverage), and component-level ablation. Because this work is reported at the design and prototype-development stage, Section 5 presents an illustrative walkthrough of the evaluation framework using representative example outputs rather than final experimental results; the complete quantitative evaluation is scheduled for completion at the end of the ten-week project timeline described in Section 4. The proposed system is intended as a research-support tool that augments, rather than replaces, human legal judgment, and is positioned to reduce unsupported claims and improve citation traceability relative to keyword-only legal search engines and standalone LLMs.

Keywords

Legal Natural Language Processing; Retrieval-Augmented Generation; Hybrid Retrieval; Court Judgment Question Answering; Explainable AI; Faithfulness Evaluation; Cross-Encoder Reranking; Legal Information Retrieval

1. INTRODUCTION

1.1 Background

Legal research depends on the ability to locate, interpret, and cite relevant precedent quickly and accurately. Court judgments, however, are long, procedurally structured, and rich in cross-references to statutes and prior cases, which makes them difficult to search using conventional keyword or Boolean methods. At the same time, general-purpose large language models have shown strong fluency in answering natural-language questions but are known to generate confident-sounding statements that are not actually grounded in any source document, a failure mode that is unacceptable in a legal context where every claim must be traceable to an authoritative text. Text analytics techniques — including information retrieval, sentence embeddings, semantic similarity, reranking, and retrieval-augmented generation — offer a way to combine the recall benefits of statistical search with the language-understanding benefits of neural models, while constraining generation to retrieved evidence.

1.2 Problem Statement

Court judgments are long, citation-heavy, and domain-specific documents. Keyword search can miss semantically related precedents because it depends on exact term overlap rather than legal meaning, while standalone LLMs used for legal question answering may produce unsupported or hallucinated legal claims because they generate text from parametric knowledge rather than from verifiable source passages. There is therefore a need for a system that retrieves legally relevant passages with high recall and precision, and that generates answers whose claims are explicitly and verifiably linked to retrieved evidence, with the ability to abstain when the evidence is insufficient.

1.3 Research Gap

Existing legal search tools and prior academic work on legal judgment prediction and legal retrieval-augmented generation tend to address retrieval, prediction, or generation in isolation, and do not consistently evaluate retrieval quality, answer correctness, and faithfulness together within a single end-to-end pipeline for court-judgment question answering (elaborated in Section 2.6).

1.4 Research Objectives

1. To build a searchable corpus of court judgments with hybrid retrieval — BM25, dense embeddings, and metadata filters — refined by cross-encoder reranking.
2. To develop legal-aware retrieval that combines dense and lexical signals and attaches case, paragraph/page, and citation-level evidence to every retrieved item.
3. To generate concise answers grounded in retrieved passages, and to evaluate retrieval quality, answer quality, and faithfulness of the generated answers.
4. To demonstrate a working prototype for legal research assistance — explicitly positioned as a decision-support aid rather than an autonomous legal decision-making system.

1.5 Research Questions

- RQ1: Does hybrid retrieval (lexical + dense + metadata filtering, refined by reranking) retrieve relevant precedents and passages more effectively than keyword-only or dense-only retrieval?
- RQ2: Does legal-structure-aware chunking (around issues, holdings, statutes, and cited precedents) improve retrieval precision compared to fixed-size chunking?

- RQ3: Does evidence-first, citation-linked generation reduce unsupported or hallucinated claims relative to unconstrained LLM generation?
- RQ4: Which pipeline component (lexical retrieval, dense retrieval, or reranking) contributes most to overall answer quality and faithfulness?
- RQ5: Is the resulting system practical for research use in terms of latency, citation coverage, and answer length?

1.6 Contributions

- Innovation: A hybrid retrieval design combining BM25/keyword search, dense embeddings, metadata filters, and cross-encoder reranking for the court-judgment domain.
- Creativity: Legal-structure-aware chunking that segments judgments around case sections, issues, holdings, statutes, and cited precedents rather than using fixed-size windows.
- Novelty: An evidence-first answer-generation strategy in which each claim is linked to retrieved passages, and unsupported claims trigger a low-confidence or abstain response instead of a fabricated answer.
- Research contribution: A multi-dimensional evaluation matrix that separately measures Recall@k / MRR, citation correctness, groundedness (faithfulness), answer relevance, and latency, rather than relying on a single aggregate score.

1.7 Organization of the Paper

The remainder of this paper is organized as follows. Section 2 reviews existing legal search approaches, machine learning and deep learning approaches to legal text analytics, and explainability techniques relevant to retrieval-augmented generation, and positions the identified research gap. Section 3 presents the proposed methodology, including the overall framework, dataset, preprocessing, feature engineering, the proposed hybrid retrieval-and-generation model, baseline models, the explainability method, and the mathematical formulation of scoring and evaluation. Section 4 describes the planned experimental setup. Section 5 presents an illustrative evaluation walkthrough. Section 6 discusses the interpretation and implications of the findings. Section 7 states the limitations of the current stage of work, and Section 8 concludes with directions for future work.

2. RELATED WORK

2.1 Existing Approaches

Most operational legal research platforms rely on keyword or Boolean search over full judgment texts. These systems return a ranked list of whole documents or fixed pages that the user must read individually to locate the relevant holding, and they provide no confidence signal about whether a returned document is actually relevant to the query. Because they match on surface-level terms rather than legal meaning, they can miss precedents that are conceptually related but lexically dissimilar, and they can split holdings and exceptions across page boundaries.

2.2 Machine Learning Approaches

Zhang and Dou (2023) show that incorporating historical case retrieval improves the realism of legal judgment prediction, supporting precedent retrieval as a first-class component of a legal reasoning

pipeline rather than an auxiliary step. Traditional statistical retrieval methods such as BM25 (Robertson & Zaragoza, 2009) remain a strong lexical baseline for legal text, particularly for exact statutory or citation matches, and are commonly combined with machine-learned ranking methods to improve precision.

2.3 Deep Learning Approaches

Neural approaches to retrieval and generation have substantially improved semantic matching over purely lexical methods. Dense Passage Retrieval (Karpukhin et al., 2020) and Sentence-BERT (Reimers & Gurevych, 2019) enable semantic similarity search using learned embeddings rather than exact term overlap, and cross-encoder re-ranking with transformer models such as BERT (Devlin et al., 2019; Nogueira & Cho, 2019) has been shown to substantially improve the precision of an initial retrieval candidate set. Retrieval-Augmented Generation (Lewis et al., 2020) formalizes the combination of a retriever with a generator so that language-model outputs are conditioned on retrieved evidence rather than generated purely from parametric knowledge, and domain-adapted legal language models such as LEGAL-BERT (Chalkidis et al., 2020) and Lawformer (Xiao et al., 2021) demonstrate that legal-domain pretraining improves downstream legal NLP tasks. Nigam et al. (2024) report that adding legal information improves legal judgment prediction, while noting that reliability at an expert-usable level remains an open problem — directly motivating the evidence-grounding approach adopted in this paper.

2.4 Explainability / Existing Techniques

Explainability in this domain concerns whether a generated legal answer can be traced back to a specific, verifiable source. Pipitone and Houir Alami (2024), through the LegalBench-RAG benchmark, find that precise, narrowly scoped passage snippets are preferable to large, imprecise chunks for legal RAG, which motivates fine-grained, legally structured chunking rather than fixed-size windows. Model-agnostic explanation techniques such as LIME (Ribeiro, Singh, & Guestrin, 2016) and SHAP (Lundberg & Lee, 2017) provide general-purpose feature-attribution explanations, but for generative legal question answering, claim-level citation attachment and entailment-style faithfulness scoring (Shuster et al., 2021; Ji et al., 2023) are more directly applicable because they explain individual generated claims rather than a single model prediction. Nigam et al. (2025), in NyayaRAG, combine realistic case context with precedent retrieval for legal judgment prediction under the Indian common-law system, motivating an India-relevant retrieval-and-explanation pipeline design.

2.5 Critical Comparison

Table 1 summarizes how the proposed system differs from existing keyword-based legal search platforms across the dimensions most relevant to evidence-grounded legal question answering.

Dimension	Existing Legal Search Websites	Proposed RAG Assistant
Search method	Keyword / Boolean search over full judgments; can miss semantically related precedents.	Hybrid retrieval — BM25 + dense embeddings + metadata filters, refined by cross-encoder reranking.
Document handling	Returns whole judgments or fixed pages; holdings and exceptions can be split apart.	Legal-structure-aware chunking around issues, holdings, statutes, and precedents.
Answer format	A ranked list of documents; the user must read each judgment to find the answer.	A concise, generated answer grounded directly in the retrieved passages.

Dimension	Existing Legal Search Websites	Proposed RAG Assistant
Evidence & citations	No claim-level links; citations must be located manually inside the judgment.	Every claim is linked to its case, paragraph/page, and citation.
Reliability check	No confidence signal; results are shown regardless of relevance.	Faithfulness check with a low-confidence / abstain response when evidence is weak.

Table 1. Existing keyword-based legal search versus the proposed hybrid RAG assistant.

Table 2 summarizes the literature reviewed and its implication for the design choices made in Section 3.

Work	Key Finding	Implication for This Project
Zhang & Dou (2023)	Historical case retrieval improves realism of legal judgment prediction.	Treat precedent retrieval as a first-class pipeline component.
Nigam et al. (2024)	Additional legal information improves prediction; expert-level reliability remains open.	Ground responses in statutes/precedents; avoid overclaiming.
Pipitone & Houir Alami (2024)	Precise snippets outperform large, imprecise chunks in legal RAG.	Optimize passage-level retrieval and legal-structure-aware chunking.
Nigam et al. (2025)	Combines realistic case context with precedent retrieval.	Motivates an India-focused retrieval-plus-explanation pipeline.
Wiggers & Hartz (2024)	Proposes measures of legal relevance specific to RAG applications.	Adopt legal-relevance-aware evaluation, not only generic IR metrics.
Magesh et al. (2024)	Leading AI legal research tools still produce unsupported or incorrect claims.	Motivates an explicit faithfulness check and abstain mechanism.

Table 2. Literature survey and design implications.

2.6 Research Gap

- Retrieval gap: Generic semantic search may retrieve topically similar but legally irrelevant text, without regard to legal hierarchy or temporal validity.
- Chunking gap: Fixed-size chunks can split holdings, exceptions, citations, and procedural context across boundaries.
- Evidence gap: A fluent answer is not sufficient; users need claim-level provenance and direct judgment evidence.
- Evaluation gap: Legal RAG requires separate measurement of retrieval relevance, answer correctness, and faithfulness, rather than a single blended score.
- Scope gap: Many prior studies address prediction or retrieval in isolation; an end-to-end court-judgment assistant must connect ingestion, retrieval, and grounded response generation.

3. PROPOSED METHODOLOGY

3.1 Overall Framework

The proposed system follows a six-stage end-to-end pipeline for court-judgment question answering: (1) Corpus — ingest judgments and associated metadata; (2) NLP Preparation — OCR and text cleaning; (3) Indexing — build a BM25 lexical index and a dense vector index; (4) Retrieval — combine hybrid top-k retrieval with cross-encoder reranking; (5) Generation — produce an answer with an LLM conditioned on the retrieved evidence; and (6) Verification — attach citations and run a faithfulness check before the answer is returned to the user. Given a user query such as "Which precedents support X?", the system returns an answer together with citations and a confidence indicator, rather than a bare document list.

3.2 Dataset

The corpus consists of publicly accessible court-judgment repositories or institution-provided judgment datasets, together with associated case metadata (court, date, jurisdiction, case type, and citation identifiers). To control scope during initial development, the corpus is deliberately limited to a single jurisdiction or court before being extended, which keeps legal terminology and citation conventions consistent within the initial corpus.

3.3 Data Preprocessing

Raw judgments are converted to clean text through OCR (for scanned or image-based sources) and text-cleaning steps that normalize whitespace, remove artefacts such as page headers/footers and stamps, and standardize citation formatting so that case and statute references can be reliably parsed downstream.

3.4 Feature Engineering

Rather than splitting judgments into fixed-size, arbitrary windows, the pipeline performs legal-structure-aware chunking that segments each judgment around its constituent sections — facts, issues, arguments, holdings, statutes cited, and precedents cited — so that a holding and its qualifying exceptions are not separated across chunk boundaries. Each chunk is enriched with structured metadata (case name, court, date, paragraph/page number, and citation identifiers) that is carried through indexing and retrieval so that every retrieved chunk can be traced back to its exact source location.

3.5 Feature Selection

The retrieval stage combines two complementary relevance signals — a lexical signal (BM25 term-weighted score) and a semantic signal (dense embedding cosine similarity) — together with hard metadata filters (for example, restricting by court or date range) applied prior to scoring. The relative weighting of the lexical and dense signals, and the number of candidates passed to the reranker, are treated as tunable hyperparameters (Section 4.3) rather than fixed a priori, so that the contribution of each signal can be evaluated empirically (Section 5.5, Section 5.7).